

T3 UV and EFT Lagrangians

EFT order: 5; loop order: 1.

$$T3 - d3 - d5 - F4, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (5, 1), \quad F = (4, \frac{1}{2})$$

$$\begin{aligned}
\text{Full UV Lagrangian} & - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2} - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_s^2} \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{N_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\
& + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1}) \\
& + i \overline{u_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_1}) \\
& + i \overline{\ell_{a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_1}) \\
& - \overline{N_{j_1, r_1}} N_{j_1, r_1} M_F \\
& + \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1, j_1} y_{1, i_1, r_1} \\
& - \overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2} \\
& - \overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}}_{a_1, j_1, j_1} \\
& - \overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{\epsilon}_{a_1, a_2} \\
& - \overline{Y_{d, i_1, i_2}} d_{\alpha_1, i_2} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e, i_1, i_2}} \bar{e}_{i_2} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& - \overline{q_{\alpha_1, a_2, i_1}} P_R u_{\alpha_1, i_2} Y_{u, i_1, i_2} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\
& + H_{a_1} \mu^2 H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& - M_{S1}^2 \text{NewScalar1}_{j_1} \text{NewScalar1}_{j_1}^\dagger \\
& + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \overline{\text{T3Y1CG}}_{a_1, j_1, j_1} \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \lambda \overline{\text{T3}} \text{NewScalar2}_{j_1} \overline{\text{T3MixCG}}_{a_1, a_2, j_1, j_1} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\
& - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \\
& + \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\
& + H_{a_1} H_{a_2} \text{NewScalar1}_{j_1} \text{T3MixCG}_{a_1, a_2, j_1, j_1} \lambda \text{T3} \text{NewScalar2}_{j_1}^\dagger \\
& + \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& +\text{lambdaH2Inv2}H_{a_2}\text{NewScalar2}_{j_2}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger\text{NewScalar2}_{j_1}^\dagger \\
& +\text{lambda12Inv1NewScalar1}_{j_2}\text{NewScalar2}_{j_2}\text{T3CrossScalarCGInv1}_{j_1,j_2,j_1,j_2}\text{NewScalar1}_{j_1}^\dagger\text{NewScalar2}_{j_1}^\dagger \\
& +\frac{1}{2}\text{lambdaS2Inv1NewScalar2}_{j_3}\text{NewScalar2}_{j_4}\text{T3Scalar2SelfCGInv1}_{j_1,j_3,j_2,j_4}\text{NewScalar2}_{j_1}^\dagger\text{NewScalar2}_{j_2}^\dagger \\
& +\frac{1}{2}\text{lambdaS2Inv2NewScalar2}_{j_3}\text{NewScalar2}_{j_4}\text{T3Scalar2SelfCGInv2}_{j_1,j_3,j_2,j_4}\text{NewScalar2}_{j_1}^\dagger\text{NewScalar2}_{j_2}^\dagger \\
& +\frac{1}{2}\text{lambdaS2Inv3NewScalar2}_{j_3}\text{NewScalar2}_{j_4}\text{T3Scalar2SelfCGInv3}_{j_1,j_3,j_2,j_4}\text{NewScalar2}_{j_1}^\dagger\text{NewScalar2}_{j_2}^\dagger \\
& +D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1}(\text{NewScalar1}_{j_1}) \\
& +D_{\mu_1}(\text{NewScalar2}_{j_1})^\dagger D_{\mu_1}(\text{NewScalar2}_{j_1})
\end{aligned}$$

$$\begin{aligned}
& \text{Matched EFT Lagrangian} - \frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{2\left(M_F^2-M_{S_1}^2\right)^2} \\
& - \frac{\sqrt{\frac{5}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{2\left(M_F^2-M_{S_2}^2\right)^2} \\
& - \frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{2\left(M_F^2-M_{S_1}^2\right)^2} \\
& - \frac{\sqrt{\frac{5}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger M_F^4}{2\left(M_F^2-M_{S_2}^2\right)^2} \\
& - \frac{2\sqrt{\frac{5}{3}}\hbar\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{\ell_{a_4,i_2}}{}^T C P_L\ell_{a_2,i_1}H_{a_1}H_{a_3}M_{S_2}^2\lambda_{\text{T}3}\bar{e}_{a_1,a_2}\bar{e}_{a_3,a_4}M_F^3}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\
& + \frac{2\sqrt{\frac{5}{3}}\hbar\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{\ell_{a_4,i_2}}{}^T C P_L\ell_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2\lambda_{\text{T}3}\bar{e}_{a_1,a_2}\bar{e}_{a_3,a_4}M_F^3}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\
& - \frac{2\sqrt{\frac{5}{3}}\hbar\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{\text{T}3}}\overline{\ell_{a_3,i_1}}C P_R\overline{\ell_{a_4,i_2}}{}^T M_{S_2}^2 y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\
& + \frac{2\sqrt{\frac{5}{3}}\hbar\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{\lambda_{\text{T}3}}\overline{\ell_{a_3,i_1}}C P_R\overline{\ell_{a_4,i_2}}{}^T M_{S_1}^2 y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\
& + \frac{\sqrt{\frac{3}{2}}\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{M_F^2-M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{M_F^2-M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}}\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2 y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)^2} \\
& + \frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2 y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{\left(M_F^2-M_{S_1}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_3,r_1} \bar{\ell}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{2,i_3,r_1} \bar{\ell}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_3,r_1} \bar{\ell}_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{2,i_3,r_1} \bar{\ell}_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar y_{1,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar y_{1,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{2,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \bar{y}_{2,i_1,r_1} \bar{Y}_{e,i_3,i_2} \bar{e}_{i_2} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \lambda_{T3} \ell_{a_3,i_1} C P_R \bar{\ell}_{a_4,i_2}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \lambda_{T3} \ell_{a_3,i_1} C P_R \bar{\ell}_{a_4,i_2}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}
\end{aligned}$$

$$\begin{aligned}
& -\frac{17hB_{\mu_1\mu_2}^2}{12\epsilon} \\
& -\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right)B_{\mu_1\mu_2}^2 \\
& -\frac{5}{12}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right)B_{\mu_1\mu_2}^2 \\
& -\frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2} \\
& -\frac{6\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{\epsilon} \\
& -5\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{1}{6}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{5}{6}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_2}) \\
& +i\overline{e_{i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_2}) \\
& +i\overline{q_{\alpha_1,a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_2}) \\
& +i\overline{u_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_2}) \\
& +i\overline{\ell_{a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_2}) \\
& -\overline{q_{\alpha_1,a_1,i_2}}P_Rd_{\alpha_1,i_1}H_{a_1}Y_{d,i_2,i_1} \\
& -\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}Y_{e,i_2,i_1} \\
& +\frac{\sqrt{\frac{3}{2}}\hbar y_{1,i_3,r_1}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\epsilon} \\
& -\frac{1}{4}\sqrt{\frac{3}{2}}\hbar y_{1,i_3,r_1}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\sqrt{\frac{3}{2}}\hbar y_{1,i_3,r_1}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}\hbar y_{1,i_3,r_1}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S1}^2\right)^2} \\
& -\frac{\sqrt{\frac{3}{2}}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S1}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{\frac{5}{2}} h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{5}{2}} h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}} \\
& - \frac{\sqrt{\frac{5}{2}} h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{M_F^2 - M_{S_2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} h \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{M_F^2 - M_{S_2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} h \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \overline{Y_{u,i_1,i_2} u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{e}_{a_1,a_2} \\
& - \overline{Y_{d,i_1,i_2} d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e,i_1,i_2} e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& + C_{H_2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{\sqrt{\frac{3}{2}} h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{3}{2}} h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{\frac{3}{2}} h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} h \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{\frac{3}{2}} h \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{5}{2}} h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{\frac{5}{2}} h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} h \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_2}^2}
\end{aligned}$$

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$$\begin{aligned}
& - \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)^2} \\
& - \overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{3 \text{lambdaH1Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2 \epsilon} \\
& + \frac{15 \text{lambdaH2Inv} 1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14 \epsilon} \\
& + \frac{95 \text{lambdaH2Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42 \epsilon} \\
& + \frac{10 \sqrt{\frac{2}{3}} \text{lambdaH2Inv} 1 \text{lambdaH2Inv} 2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7 \epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv} 2^2 \hbar \log \left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{15}{14} \text{lambdaH2Inv} 1^2 \hbar \log \left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{95}{42} \text{lambdaH2Inv} 2^2 \hbar \log \left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{10}{7} \sqrt{\frac{2}{3}} \text{lambdaH2Inv} 1 \text{lambdaH2Inv} 2 \hbar \log \left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4 g_Y^2}
\end{aligned}$$
